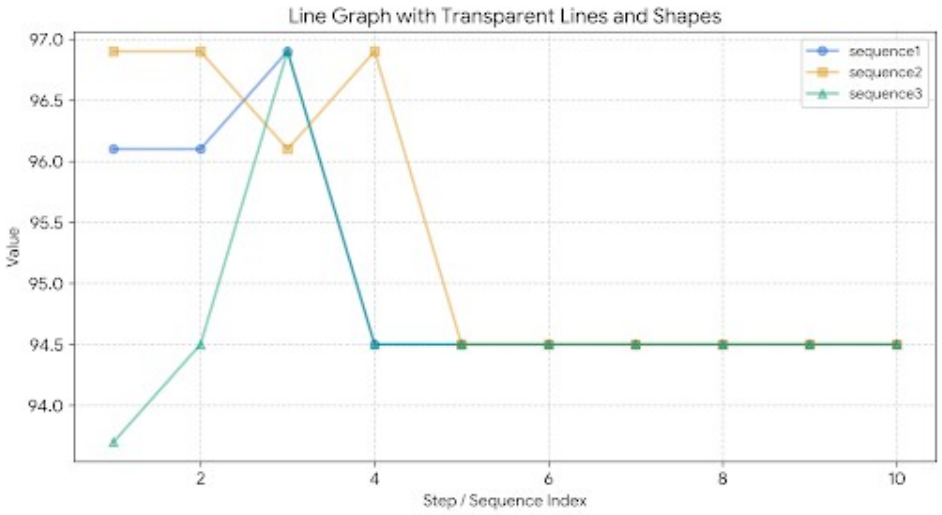
Transparent Line Graph Generation

https://gemini.google.com/app/5d6033ad1e3d06e5

User prompt: make a line graph with legend and transparent lines with shapes at each point



I have successfully processed your dataset (`data.csv`) and generated the requested line graph.

Key Features of the Graph:

- **Transparent Lines:** The lines have a transparency level (`alpha = 0.4`) to emphasize the paths clearly.
- **Shapes (Markers):** Distinct markers (`o`, `s`, `^`) are placed at each data point.
- **Legend & Grid:** A clear legend identifying each sequence and a dashed grid for readability have been included.

User prompt: give python code, seaborn

Response: Here is the complete Python code using `seaborn` and `matplotlib` to generate the exact line graph with transparent lines, distinct markers at each point, and a legend:

```
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

# Load the dataset
df = pd.read_csv("data.csv")

# Reshape the data from wide to long format for Seaborn
df_melted = df.melt(
    id_vars=["sequence"], var_name="step", value_name="value"
)
df_melted["step"] = df_melted["step"].astype(int)
df_melted["value"] = df_melted["value"].astype(float)
```

```
# Set the visualization style
sns.set_theme(style="whitegrid")
plt.figure(figsize=(9, 5))

# Create the line plot with markers
ax = sns.lineplot(
    data=df_melted,
    x="step",
    y="value",
    hue="sequence",
    style="sequence",
    markers=["o", "s", "^"],
    linewidth=2,
    markersize=6,
    alpha=0.4,  # Sets transparency for both lines and markers
)

# Customize plot details
plt.title("Line Graph with Transparent Lines and Shapes", fontsize=14)
plt.xlabel("Step / Sequence Index", fontsize=12)
plt.ylabel("Value", fontsize=12)
plt.legend(title="Sequence")
plt.tight_layout()

# Save and show the plot
plt.savefig("line_graph_seaborn.png", dpi=300)
plt.show()
```
